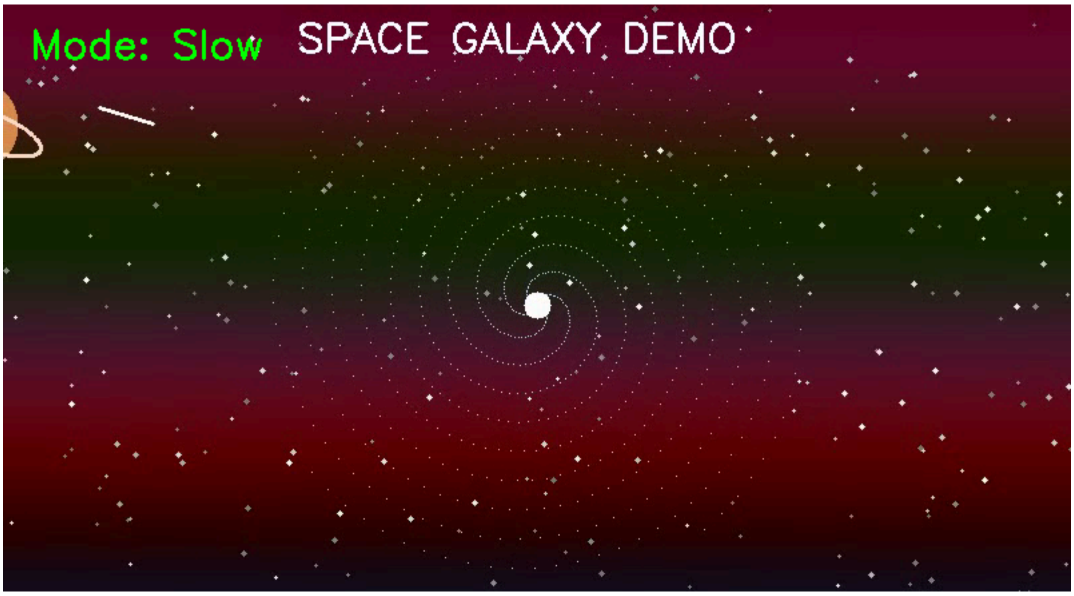
```
import cv2
import time
from google.colab.patches import cv2_imshow
from IPython.display import clear_output

def process_and_display_video(video_path, speed="normal"):
    cap = cv2.VideoCapture(video_path)

    if not cap.isOpened():
        print("Error: Could not open video file.")
        return

    fps = cap.get(cv2.CAP_PROP_FPS)
    if fps == 0 or fps is None:
        fps = 30

    normal_delay = 1.0 / fps

    if speed == "slow":
        delay = normal_delay * 2.5
    elif speed == "fast":
        delay = normal_delay / 3.0
    else:
        delay = normal_delay
    print(f"Playing video in **{speed.upper()}** mode (FPS: {fps:.2f})...")

    while cap.isOpened():
        ret, frame = cap.read()


        if not ret:
            print("End of video stream.")
            break


        cv2.putText(
            frame,
            f"Mode: {speed.capitalize()}",
            (20, 40),
            cv2.FONT_HERSHEY_SIMPLEX,
            1,
            (0, 255, 0),
            2
        )

        clear_output(wait=True)

        cv2_imshow(frame)


        time.sleep(delay)

    cap.release()

video_file = "video_path.mp4"
process_and_display_video(video_file, speed="slow")
```


Mode: Slow
SPACE GALAXY DEMO